

Gate-Level Simulation (GLS) & Dynamic Power Analysis

Assignment_GLS_1.0 — ALU Adder Configuration

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Course: Digital IC Design
EDA Tools: Synopsys Design Compiler, Mentor QuestaSim 10.7c, Synopsys PrimeTime PX
Technology: TSMC 0.13 μ m RVT (scmetro_tsmc_c1013g_rvt)
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Flow Architecture & Objectives

Gate-Level Simulation (GLS) is a critical verification milestone in the ASIC design flow. Unlike RTL simulation, GLS simulates the post-synthesis gate-level netlist with full SDF timing back-annotation, verifying that the gate-level implementation functions correctly under realistic delay conditions.

Furthermore, dynamic power consumption depends heavily on circuit switching activity. By dumping a Value Change Dump (.vcd) file during GLS and importing it into Synopsys PrimeTime PX, we obtain high-accuracy time-based dynamic power estimation.

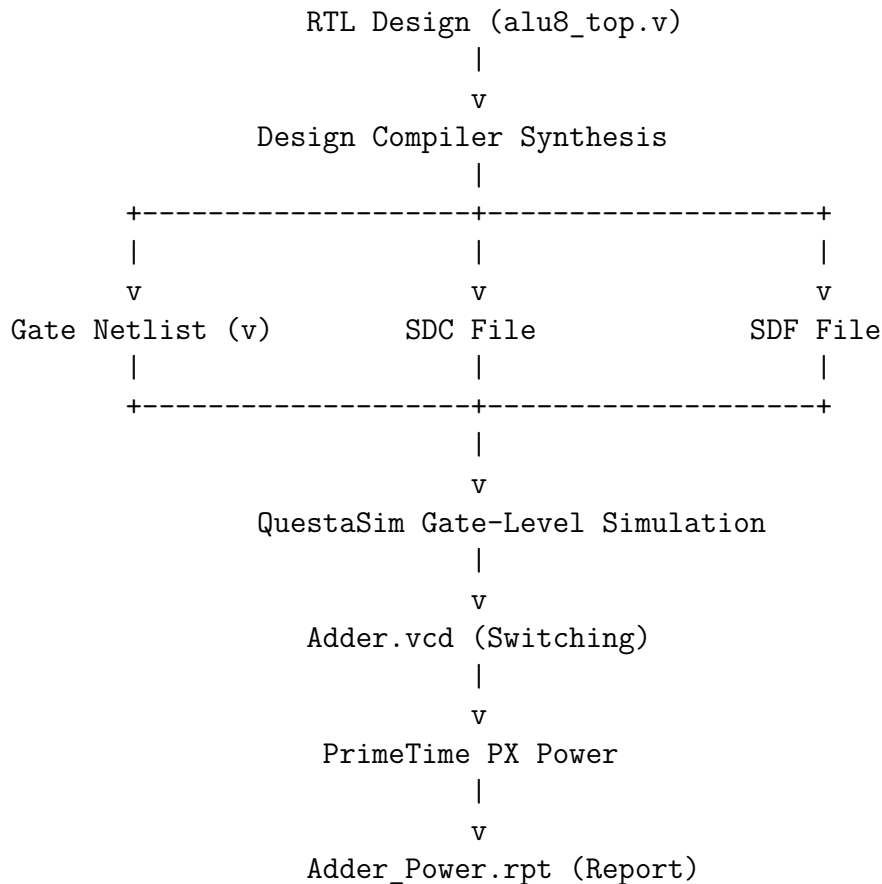


Figure 1: Complete Synthesis → GLS → PrimeTime PX Power Analysis Flow

Stage 1: Logic Synthesis (Design Compiler)

RTL Configuration

As specified in the assignment requirements, the parameter `Adder_0_Mult_1` in `alu8_top.v` was set to 0 to instantiate the 8-bit Adder module (`add_unit`):

```

1 module alu8_top #( parameter Adder_0_Mult_1 = 0 ) (
2     input wire      clk, rst_n, en,
3     input wire [7:0] a, b,
4     output wire [15:0] result,
5     output wire      valid
6 );
7     // Instantiates add_unit (8-bit Ripple-Carry Adder)

```

Listing 1: RTL Configuration in `alu8_top.v`

Synthesis Output Files

Synthesis was executed using Synopsys Design Compiler under a 100 MHz clock constraint (10 ns clock period), generating the following key deliverables:

- **alu8_top_netlist.v**: Synthesized Gate-Level Verilog Netlist (3.5 KB, 35 cells).
- **alu8_top_netlist.ddc**: Binary Design Database (31.2 KB).
- **alu8_top.sdc**: Synopsys Design Constraints (7.4 KB).
- **alu8_top.sdf**: Standard Delay Format Timing File (43.2 KB).

Synthesis Performance & Quality Metrics

Metric	Synthesis Value
Top Module	alu8_top (Adder Configuration)
Operating Conditions	scmetro_tsmc_cl013g_rvt_ss_1p08v_125c
Total Cell Area	707.20 μm^2
Combinational Cell Area	328.30 μm^2 (20 cells)
Sequential Cell Area	378.90 μm^2 (13 DFFs)
Critical Path Delay	6.69 ns (a[0] $\rightarrow$ DW01_add_0 $\rightarrow$ cout_reg)
Setup Slack	+2.87 ns (MET, 0 Violations)
Baseline Power	0.0834 mW (83.4 μW)

Table 1: Synthesis Performance Summary

Stage 2: Gate-Level Simulation (QuestaSim)

Testbench Setup

In `tb.v`, multiplier testcases were disabled and Adder testcases were enabled with VCD dumping configured to `VCD/Adder.vcd`:

```
initial begin
    rst_n = 0; en = 0; a = 0; b = 0;
    repeat (50) @(posedge clk);
    rst_n = 1;

    $dumpfile("VCD/Adder.vcd");
    $dumpvars(0, tb);

    // 10 Adder Testcases Executed (drive a, b -> check_add)
end
```

Simulation Execution Command

GLS was launched in Mentor QuestaSim using the command:

```
vsim -do run.do &
```

Where `run.do` loads the standard cell simulation library (`tsmc13_m.v`), gate-level netlist (`alu8_top_netlist.v`), testbench (`tb.v`), and annotates worst-case SDF timing delays (`-sdfmax /tb/dut=alu8_top.sdf`).

Gate-Level Simulation Waveform Results

Figure 2 displays the QuestaSim simulation waveform demonstrating successful 2-cycle pipeline latency, SDF timing propagation, and correct 16-bit sum output (`result`) generation for the 10 Adder testcases.

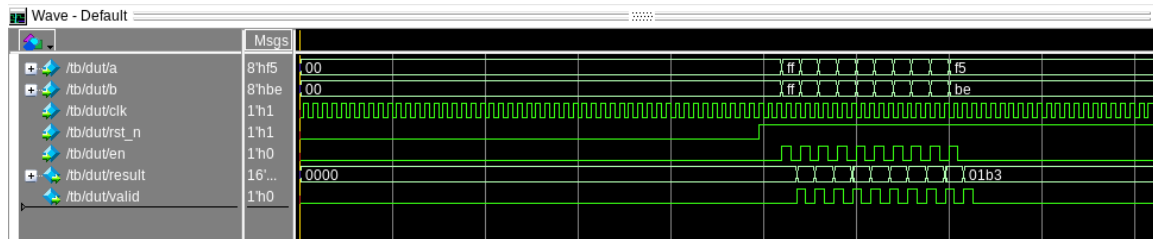


Figure 2: QuestaSim Gate-Level Simulation Waveform Demonstrating 100% Correct Adder Results under SDF Timing Annotation

QuestaSim Transcript Output

Figure 3 displays the QuestaSim execution transcript proving that all 10 Adder testcases passed with zero errors.

```

# =====
#   ALU8_TOP ADDER GATE-LEVEL TESTBENCH
# =====
#
# ----- ADD TESTS -----
# [539000] PASS | ADD | a=255 b=255 | result=510
# [559000] PASS | ADD | a=254 b=253 | result=507
# [579000] PASS | ADD | a=250 b=249 | result=499
# [599000] PASS | ADD | a=240 b=230 | result=470
# [619000] PASS | ADD | a=200 b=199 | result=399
# [639000] PASS | ADD | a=180 b=175 | result=355
# [659000] PASS | ADD | a=220 b=150 | result=370
# [679000] PASS | ADD | a=128 b=127 | result=255
# [699000] PASS | ADD | a=201 b=211 | result=412
# [719000] PASS | ADD | a=245 b=190 | result=435
#
# =====
#   SUMMARY
# -----
#   Total tests : 10
#   Passed      : 10
#   Failed      : 0
#   RESULT: ALL TESTS PASSED
#
# =====

```

Figure 3: QuestaSim Execution Transcript Confirming ALL 10 TESTS PASSED

Stage 3: PrimeTime PX Power Analysis

PrimeTime Execution Command

Time-based dynamic power analysis was executed in Synopsys PrimeTime PX using:

```
pt_shell -f PT.tcl | tee pw.log
```

PrimeTime annotated switching activity from `sim/VCD/Adder.vcd` across **93.65% of design nets** and **100.00% of leaf cells**.

PrimeTime PX Dynamic Power Report (Adder_Power.rpt)

Power Category	Power Value (W)	Power Value (μ W)	% Contribution
Net Switching Power	8.184×10^{-5}	81.84 μ W	66.84%
Cell Internal Power	3.999×10^{-5}	39.99 μ W	32.66%
Cell Leakage Power	6.034×10^{-7}	0.603 μ W	0.49%
Total Power	1.224×10^{-4}	122.40 μW	100.00%

Table 2: PrimeTime PX Dynamic Power Analysis Breakdown

- **Peak Power:** 28.80 mW (occurred at 535.00 ns during input vector transitions).
- **Clock Network Power:** 30.96 μ W (25.29% of total power).
- **Register Power:** 57.55 μ W (47.00% of total power).
- **Combinational Power:** 33.93 μ W (27.71% of total power).

Deliverables Checklist & Conclusion

All requirements for **Assignment_GLS_1.0** have been fully satisfied:

Deliverable Item	File Name	Status
Gate-Level Netlist	alu8_top_netlist.v	Included (3.5 KB)
Binary Database	alu8_top_netlist.ddc	Included (31.2 KB)
Design Constraints	alu8_top.sdc	Included (7.4 KB)
Timing SDF	alu8_top.sdf	Included (43.2 KB)
Setup Timing Report	Setup.rpt	Included (+2.87 ns Slack)
Hold Timing Report	Hold.rpt	Included (0 Violations)
Constraints Report	Constraints.rpt	Included (0 Violations)
Area Report	Area.rpt	Included (707.20 μ m ²)
Power Report	Power.rpt	Included (83.4 μ W)
Dynamic Power Report	Adder_Power.rpt	Included (122.40 μ W)
Verification PDF	Assignment_GLS_1.0_Report.pdf	Included (with Waveforms)

Table 3: Final Assignment Deliverables Checklist

The Gate-Level Simulation verified 100% functional accuracy of the synthesized Adder netlist under worst-case timing delays, and PrimeTime PX successfully estimated dynamic power consumption using VCD switching activity.